

LUNA-01

MISSION CONTROL

Game Design Document (GDD)

NASA Space Apps Challenge 2026 - Space Mission Design Game

Project vision

A short, replayable, browser-based 2D lunar mission simulation where the player designs and manages one robotic science mission and sees how engineering trade-offs change the outcome.

Target MVP playtime: 5-7 minutes | Development target: 5 days

1. Project Identity

Item	Specification
Working Title	LUNA-01: Mission Control
Genre	Educational Strategy / Simulation / Decision-Making Game
Event	NASA Space Apps Challenge 2026 - Space Mission Design Game
Platform	Web Browser
Rendering	HTML5 Canvas 2D
Interface	HTML/CSS with Canvas-based visual scene
Programming	Vanilla JavaScript
Data	NASA lunar CSV / locally curated structured data
Audio	Web Audio API / procedurally generated audio
Backend	None for MVP
Database	None for MVP
Game Engine	None
Visual Assets	Code-generated; no external image assets for the core game
Audio Assets	Code-generated; no external audio files for the core game
Intended Playtime	Approximately 5-7 minutes
Development Constraint	Small, polished, playable MVP suitable for a five-day first submission

2. Project Vision

LUNA-01 is a short interactive game in which the player becomes the Mission Director of a single robotic lunar science mission.

The player must define a scientific objective, design a spacecraft within constraints, select a launch vehicle and scientific instrument, make a simplified lunar orbital decision, perform a lunar survey using genuine NASA information, manage limited resources, transmit the collected data, and receive a final mission evaluation.

Core player lesson

There is no perfect space-mission design. Every advantage comes with a cost.

3. Challenge Alignment

The game is designed around the challenge requirement to let participants design, manage, and simulate a complete space mission while making engineering decisions, managing limited resources, and evaluating how each choice shapes mission success.

Challenge requirement	LUNA-01 implementation
Mission objectives	Player receives a lunar science objective.
Spacecraft design	Player configures the spacecraft.
Scientific instruments	Player selects one scientific instrument.
Launch vehicles	Player chooses a launch vehicle.
Budget	Mission starts with a limited budget.
Power	Spacecraft has limited power.
Mass	Design must stay within a mass limit.
Communications	Communication capability influences data transmission.
Orbital constraints	Player chooses a simplified lunar orbital strategy.
Engineering decisions	Multiple choices have benefits and costs.
Limited resources	Budget, power, fuel, mass and science are managed/derived.
Mission simulation	Mission progresses through checkpoints.
Mission management	Earlier decisions alter later mission conditions.
Evaluation	Final report explains success, partial success, or failure

4. Scope Philosophy

The MVP must remain intentionally small. The project is optimized for a five-day first submission, followed by academic exams. The first goal is a dependable and polished playable experience, not a complete aerospace simulator.

- One mission only.
- One destination: the Moon.
- One robotic spacecraft.
- Four major gameplay checkpoints.
- Short session length.
- Small set of resources and decisions.
- Small curated NASA dataset used meaningfully.
- Procedurally generated visuals.
- Procedurally generated audio.
- No backend, database, or game engine.

Explicitly out of scope for the MVP

- 3D graphics.
- Multiple missions or planets.
- Multiplayer.
- Real-time orbital mechanics.
- Detailed rocket or spacecraft physics.
- Complex AI.
- Large scientific-data processing.
- Live API dependency.
- User accounts.
- Procedural universe generation.
- External image assets.
- External audio files.

5. Core Game Concept

The player is the Mission Director. The player does not manually pilot the spacecraft. Instead, the player makes mission-level decisions and receives clear consequences.

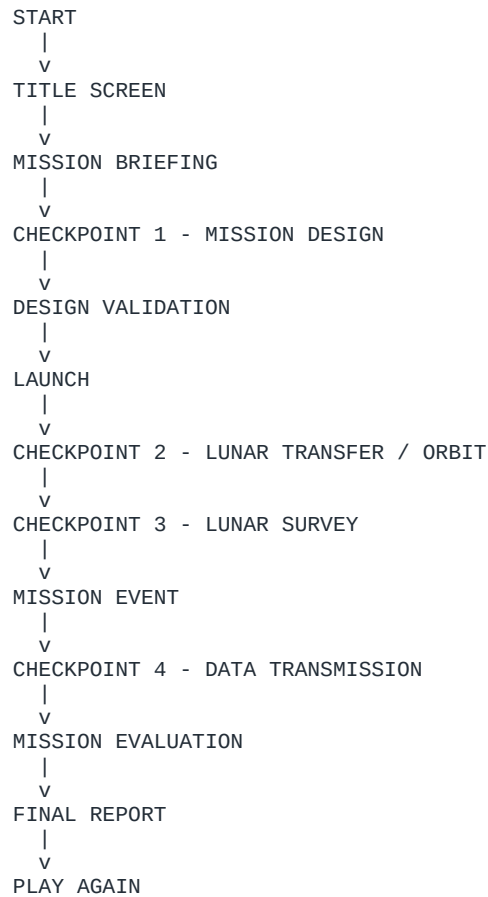
Intended feeling

The player should think: "My decisions determine whether this mission succeeds."

6. Mission Premise

Item	Definition
Mission Name	LUNA-01
Mission Type	Robotic lunar science mission
Main Objective	Reach the Moon, survey a selected lunar region, collect useful scientific data, and transmit it to Earth.
Mission Path	Earth -> Lunar Transfer -> Lunar Orbit -> Survey -> Data Transmission -> Result
Mission Duration in Game	Short, checkpoint-based experience; no simulated long-duration clock is required.

7. Overall Game Flow



8. Core Gameplay Loop

```
Observe situation
->
Review current resources
->
Choose an option
->
Apply consequences
->
Show visual/audio feedback
->
Continue to next checkpoint
```

9. Core Mission Resources

Resource	Meaning	Gameplay use
Budget	Funding available to build the mission.	Consumed by rocket, power system and instrument choices.
Power	Available spacecraft energy.	Consumed by instruments, environmental effects and communication.
Fuel	Available propulsion margin.	Consumed by travel/orbit-related actions and selected mission strategies.
Science	Scientific value collected by the mission.	Increases through instrument selection, target conditions and observation choices.

10. Derived Mission Factors

Factor	Role
Mass	Design constraint. Equipment increases mass; excess mass can make the design invalid or less efficient.
Communication Capability	Primarily determined by equipment; influences transmission performance.
Mission Risk	Derived internal value influenced by low resources and aggressive decisions. It is not a primary resource-management system.

11. Central Gameplay Rule

Every major choice must have a benefit and a cost.

Examples: better instruments increase science but consume power, mass and budget; better communication improves transmission but costs power/mass/budget; more fuel improves margin but adds mass and cost; aggressive orbit increases science potential but consumes more resources and may increase risk.

12. Resource and Decision Examples

Decision	Positive effect	Negative effect
Higher science instrument	Science increases significantly.	Power, mass and budget decrease/increase in cost terms.
High-gain communication	Communication capability increases.	Consumes budget, mass and power.
More fuel reserve	Fuel margin increases.	Mass and budget are consumed.
Aggressive lunar orbit	Science potential increases.	Fuel/power consumption and risk increase.
Deep scan	Largest science gain.	Largest power cost and possible risk increase.
Full-resolution transmission	Maximum scientific return.	Higher power demand.

13. CHECKPOINT 1 - Mission Design

The player constructs a viable mission before launch.

13.1 Mission Brief

The player receives a simple objective: survey a lunar region and transmit meaningful scientific data to Earth.

Example starting constraints (tunable during balance testing):

Budget: \$100M
 Maximum Mass: 800 kg
 Initial Power: 100%
 Initial Fuel: 100%
 Minimum Science Goal: 60

13.2 Launch Vehicle Selection

Option	Design role	Trade-off
Light Rocket	Low-cost / low-capacity	Cheap but restrictive payload capacity.
Medium Rocket	Balanced	Moderate cost and payload capacity.
Heavy Rocket	High-capacity	More capable but consumes much more budget.

13.3 Power System Selection

Option	Design role	Trade-off
Basic Solar	Low cost / low mass	Lower power reserve.
High-Capacity Solar	Higher power	Higher cost and mass.
Solar + Battery Backup	Resilience	Highest cost/mass but better protection from low-light periods.

13.4 Scientific Instrument Selection

Instrument	Primary science	Trade-off
Camera	Surface imaging	Lower cost/power; moderate science.
Spectrometer	Composition analysis	Higher science; higher power/mass/cost.
Radiation Sensor	Radiation/environment measurements	Moderate-high science; resource demand.

13.5 Design Summary Panel

MISSION DESIGN

Budget Used \$72M
 Mass 650 kg
 Power Capacity 84%
 Fuel Capacity 85%
 Communication 70
 Science Potential 68

13.6 Design Validation

Before launch, the game validates the chosen combination.

DESIGN REVIEW

Budget OK
 Mass OK
 Power OK
 Mission OK

READY FOR LAUNCH

When constraints are violated, the player returns to the design screen to modify one or more selections.

14. Launch Sequence

The launch is a short automatic simulation. No manual rocket control and no real rocket physics are required.

EARTH
 0
 |
 |
 ^
 |

ROCKET

LAUNCH SEQUENCE
[=====] 100%

LAUNCH SUCCESSFUL

The Canvas animation should use simple motion, engine glow, particles, and generated launch audio to create feedback.

15. CHECKPOINT 2 - Lunar Transfer / Orbit

The spacecraft travels toward the Moon. The Canvas shows Earth on one side, the Moon on the other, and an animated trajectory.



16. Simplified Lunar Orbital Decision

Choice	Benefits	Costs
Low Lunar Orbit	Higher scientific observation potential.	Higher fuel and power consumption.
Higher Lunar Orbit	Lower resource consumption and lower operational risk.	Lower science potential.

Simulation rule

Do not implement real orbital mechanics in the MVP. Use a transparent game abstraction that represents the concept of orbital trade-offs.

17. NASA Data Integration

The game uses the supplied NASA lunar dataset as a real input source. The dataset contains approximately 21 lunar/environmental values with units, notes, and NASA source links.

The dataset should be treated as the authoritative source for any scientific value that the game uses. Claude Code must not silently invent replacement scientific values when the supplied dataset already contains a relevant one.

18. NASA Data Categories

Category	Representative values	Possible game role
Physics	Gravity, escape velocity, size/orbital information	Simplified travel/orbit modifiers and education.
Environment	Temperature, atmosphere, oxygen, lunar day/night	Power demand, environmental explanation, difficulty.
Resources	Water and ice information	Science value and target attractiveness.
Hazards	Radiation, extreme temperature	Risk, event triggers, scientific context.

19. Key NASA Values Available to the Game

Value	Known figure from supplied dataset	Possible use
Lunar gravity	1.62 m/s ²	Simplified travel/orbit modifier; educational display.
Escape velocity	2.38 km/s	Mission-difficulty context; not a full propulsion model.
Surface temperature	Approximately 95 K to 390 K; about	Environmental challenge and

	-178 C to 117 C	power/risk effects.
Coldest measured location	Approximately 26 K; about -249 C	Extreme-environment event or educational callout.
Atmosphere	Extremely thin; supplied pressure around 3×10^{-15} bar	Environmental information.
Oxygen	No breathable oxygen	Environmental explanation; not a resource in the robotic mission.
Lunar daylight/night	About 14.8 Earth days each	Solar-power challenge / low-light event.
Water/ice	Polar ice and LCROSS-related information	Science bonus / target attractiveness.
Estimated total lunar ice	About 6 trillion kg, marked uncertain in supplied notes	Optional low-confidence information only; never present as a precise measurement.
Historical radiation	About 0.8-1.5 millirad/hour	Hazard context and/or science modifier.

20. NASA Data Presentation Rule

The game should never turn the dataset into a spreadsheet experience. NASA information should appear naturally inside the mission interface.

LUNAR ENVIRONMENT

Gravity	1.62 m/s ²
Temperature	-178 C to 117 C
Water/Ice	Potentially present
Day/Night	~14.8 days each
Radiation	0.8-1.5 millirad/hour

Each displayed value should retain its unit and source context. Where the dataset marks an estimate or uncertainty, the game should communicate that clearly.

21. Scientific Accuracy Policy

Scientific processes are intentionally simplified for gameplay. Simplified formulas and scores must be described as gameplay abstractions or approximations rather than exact NASA physics.

- Real scientific value -> display as sourced data.
- Estimate/model -> clearly label as estimate or uncertain.
- Gameplay value derived from science -> label internally as a game modifier/approximation.
- Do not claim that a mission score, risk number, or fuel cost is an official NASA measurement.

22. CHECKPOINT 3 - Lunar Survey

The spacecraft reaches the Moon and the player performs the primary scientific activity of the mission.

22.1 Canvas Scene

- Large Moon in the center or dominant side of the scene.
- Procedural craters and surface shading.
- Highlighted survey target with a pulsing marker.
- Spacecraft orbiting or hovering near the Moon.
- Scanning beam or sweeping sensor effect.
- Mission route and checkpoint indicator.

22.2 Lunar Target Information

The player sees a selected lunar region with a small amount of relevant NASA-backed information.

LUNAR SURVEY TARGET

TARGET: POLAR REGION

Mode	Science	Power	Risk
Quick Scan	Low	Low	Low
Standard Scan	Medium	Medium	Balanced
Deep Scan	High	High	Higher

- Camera + Deep Scan -> strong imaging science return with high power demand.
- Spectrometer + Deep Scan -> strong scientific-value increase with higher power demand.
- Radiation Sensor + Deep Scan -> strong environmental science return with moderate/high resource use.

Example event: POWER DEMAND INCREASE
Environmental conditions increase power consumption. The player must choose what to sacrifice.

MOON 0 ----- [SPACECRAFT] ----- 0 EARTH
 . . .
 DATA PULSES

Choice	Effect
Compressed Transmission	Lower power consumption; reduced scientific-data quality/value.
Full-Resolution Transmission	Higher power consumption; maximum scientific return.

Game Design Document

- The result should depend on the actual mission state rather than a purely random outcome.

28. Mission Result

Outcome	Meaning
SUCCESS	Primary mission objective achieved and data successfully transmitted.
PARTIAL SUCCESS	Meaningful science achieved but some performance, data, or resource objective was lost.
FAILURE	A critical mission requirement was not achieved.

29. Final Mission Report

LUNA-01 MISSION REPORT

```
MISSION STATUS      SUCCESS
SCIENCE COLLECTED   82
POWER REMAINING     38%
FUEL REMAINING      52%
BUDGET USED         79%
MISSION RISK        LOW
```

30. Decision Impact Report

The final screen should explicitly connect player decisions to mission outcomes.

YOUR KEY DECISIONS

```
Spectrometer
Science      +25
Power        -10
Mass         +20 kg
```

```
Low Lunar Orbit
Science      +10
Fuel         -12
Power        -8
```

```
Deep Scan
Science      +20
Power        -15
```

```
Full Transmission
Science      +8
Power        -10
```

31. Replayability

There is intentionally only one mission. Replayability comes from different strategies and different combinations of decisions.

Strategy	Typical approach
Science-focused	Choose stronger instruments, higher observation intensity, and accept greater resource consumption.
Safety-focused	Choose conservative orbit/power options and protect mission margins.
Efficiency-focused	Choose lower-cost equipment and prioritize resource conservation.

32. Visual Design

Visual identity

Retro-futuristic mission control: dark space, clean technical panels, vector-like geometry, procedural stars and planets, subtle glow, and restrained sci-fi motion.

33. Visual Asset Philosophy

No external image assets are required for the core experience. Every important visual object is created with code and rendered through Canvas.

Element	Implementation concept
Stars	Seeded/procedural points and particles.
Moon	Circle/gradient + procedural craters + target highlight.
Earth	Circle/gradient + simplified procedural land/cloud shapes.
Spacecraft	Polygons/rectangles/lines/circles; procedural solar panels and antenna.
Rocket	Simple geometric rocket + procedural flame.
Trajectory	Canvas lines and animated travel markers.
Survey beam	Animated line/cone/sweep effect.
Transmission	Animated pulses traveling toward Earth.
Particles	Launch exhaust, subtle ambient particles, success/warning effects.
UI icons	Canvas-drawn symbols or CSS where more practical; no image dependency.

34. Canvas Responsibilities

- Main space environment.
- Earth and Moon.
- Spacecraft and rocket.
- Orbit/trajectory visualization.
- Launch, travel, survey and transmission animations.
- Procedural stars, craters and particle effects.
- Mission visual feedback and transitions.

35. CSS Responsibilities

- Overall layout.
- Responsive sizing.
- Typography.
- Supporting interface panels.
- Buttons and cards where easier than Canvas.
- Accessibility and focus styling.

The game should still feel like a Canvas-based game rather than a standard HTML dashboard with a decorative Canvas background.

36. JavaScript Responsibilities

- Mission state.
- Game/checkpoint state machine.
- Resource calculations.
- Decision consequences.
- NASA CSV loading and parsing.
- Canvas rendering.
- Animation loop.
- Input handling.
- Audio generation.
- Final scoring and outcome calculation.
- Replay/reset logic.

37. Audio System

The MVP should not depend on external sound files. Use the Web Audio API to generate short, retro-digital sounds and a lightweight procedural music loop.

38. Procedural Audio Style

The sound should have a subtle retro / pixelated / arcade-inspired character, inspired by the team's previous game-jam approach, but still fit the mission-control theme.

Audio element	Implementation idea
Background music	Short procedural synth loop using oscillators, notes and filters.
Button click	Very short square/triangle-wave beep.
Selection confirmation	Short positive two-tone sound.
Warning	Two-tone or descending digital alert.
Launch	Rising-frequency tone + controlled noise burst.
Survey	Subtle scanning pulse.
Transmission	Repeating digital data pulses.
Success	Short melodic arpeggio.
Failure	Low descending tone.

Audio rule

Gameplay must remain fully understandable when audio is muted. Audio provides immersion and feedback; it is never the only source of information.

39. User Interface Flow

```
TITLE
->
MISSION BRIEFING
->
MISSION DESIGN
->
DESIGN REVIEW
->
LAUNCH
->
LUNAR TRANSFER
->
ORBIT DECISION
->
LUNAR SURVEY
->
MISSION EVENT
->
DATA TRANSMISSION
->
MISSION RESULT
->
FINAL REPORT
->
PLAY AGAIN
```

40. Persistent Mission HUD

```
LUNA-01

POWER    [===== - -] 82%
FUEL     [===== - -] 61%
BUDGET   [===== - -] 74%
SCIENCE  [===== - -] 58

CHECKPOINT 3 / 4
```

The HUD should make mission status visible without overwhelming the player.

41. Checkpoint Indicator

```

0 DESIGN
|
0 LAUNCH
|
0 ORBIT
|
0 SURVEY
|
0 TRANSMISSION
|
0 RESULT

```

42. Scene-by-Scene Canvas Blueprint

42.1 Scene 1 - Title

- Canvas: star field, Moon, spacecraft, title.
- Slow ambient motion.
- Button: Start Mission.

42.2 Scene 2 - Mission Briefing

- Canvas: Earth, Moon, mission route and spacecraft.
- Overlay: mission objective and starting constraints.
- Action: begin design.

42.3 Scene 3 - Mission Design

- Canvas: central spacecraft schematic.
- Controls: rocket, power system and scientific instrument.
- HUD: budget, mass, power, fuel, communication, science potential.

42.4 Scene 4 - Design Review

- Canvas: spacecraft with status highlights.
- Panel: valid/invalid constraints.
- Action: launch or redesign.

42.5 Scene 5 - Launch

- Canvas: Earth horizon, rocket, flame, exhaust particles.
- Animation: rocket departs Earth.
- Audio: procedural launch cue.

42.6 Scene 6 - Lunar Transfer

- Canvas: Earth left, Moon right, spacecraft on trajectory.
- Animation: spacecraft movement.
- HUD: current mission resources.

42.7 Scene 7 - Orbit Decision

- Canvas: Moon with two stylized orbital paths.
- Choices: Low Orbit or Higher Orbit.
- Show simplified resource consequences before confirmation.

42.8 Scene 8 - Lunar Survey

- Canvas: large Moon, target marker, spacecraft, scan beam.
- Panel: small NASA data snapshot.
- Choices: Quick, Standard, Deep Scan.

42.9 Scene 9 - Mission Event

- Canvas: warning feedback and environmental visual.
- Panel: short event description and three responses.
- Apply immediate resource changes.

42.10 Scene 10 - Transmission

- Canvas: Moon, spacecraft, Earth, animated data pulses.
- Choices: Compressed or Full-Resolution.
- Audio: data pulse sequence.

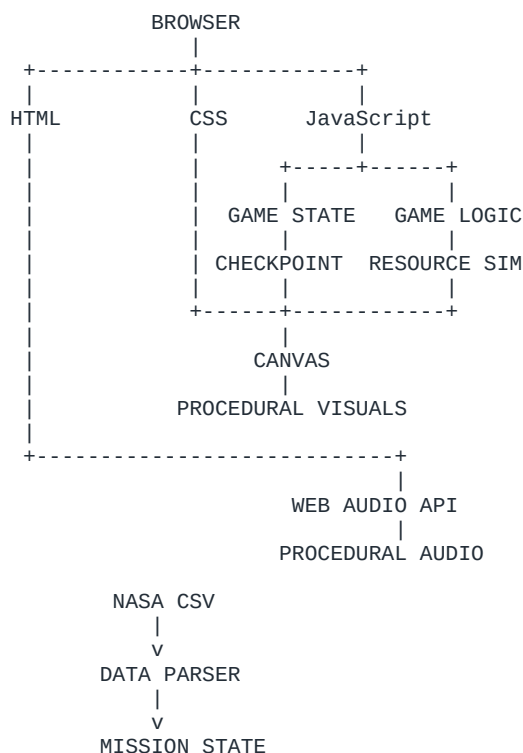
42.11 Scene 11 - Mission Result

- Canvas: completed mission route or failure state.
- Panel: outcome summary.

42.12 Scene 12 - Final Report

- Canvas: calm mission-control scene, Moon and stars.
- Panel: final stats and decision impact.
- Action: Play Again.

43. Technical Architecture



44. Suggested Project Structure

```

luna-mission-control/
|
+- index.html
+- style.css
+- app.js
+- CLAUDE.md
+- README.md
|
+- data/
|   \`- moon_environment_dataset.csv
|
\`- src/
    +- game.js
  
```

```

+- state.js
+- data.js
+- renderer.js
+- input.js
+- audio.js
+- missions.js
+- ui.js
+- simulation.js
\`- utils.js

```

The exact number of files may be reduced during implementation if a smaller structure is clearer. Avoid abstraction for its own sake.

45. Game State Model

```

MISSION STATE

missionStage
checkpoint
budget
mass
power
fuel
science
communication
risk

selectedRocket
selectedPowerSystem
selectedInstrument
selectedOrbit
selectedObservation
selectedTransmission

missionStatus

```

46. State-Based Architecture

```

TITLE
BRIEFING
DESIGN
DESIGN_REVIEW
LAUNCH
TRANSFER
ORBIT_DECISION
SURVEY
MISSION_EVENT
TRANSMISSION
RESULT
REPORT

```

Only one state is active at a time. The renderer draws the correct Canvas scene based on the current state. All decisions modify the shared mission state.

47. NASA Dataset Handling

Use the supplied CSV directly as part of the project. Preserve source links, units, and notes. Do not regenerate the dataset unless a conversion is strictly necessary for parsing.

```

data/
  moon_environment_dataset.csv

```

Recommended logical fields for the game data model: name, value, unit, category, confidence, gameUse, source. If the current CSV uses different headers, adapt to the actual file instead of creating duplicate values.

48. Claude Code Development Rules

1. Build the smallest playable version first.
2. Do not introduce features outside this GDD unless explicitly requested.
3. Do not introduce a game engine.
4. Do not introduce a backend or database.
5. Use Canvas for the main visual experience.

6. Use genuine NASA dataset values where applicable.
7. Do not invent scientific values when the supplied dataset already contains the needed value.
8. Do not implement realistic orbital mechanics in the MVP.
9. Do not rewrite working systems unnecessarily.
10. Keep the game playable after every major development stage.
11. Prefer simple, transparent formulas over opaque systems.
12. Before adding a dependency, check whether native browser APIs are sufficient.
13. Prioritize reliability, clarity, and demo quality over technical sophistication.

49. Recommended Development Sequence

Phase	Main work
1. Skeleton	HTML/CSS/Canvas shell, game loop, state manager, input.
2. Core Mission	Briefing, design, resources, validation, checkpoints.
3. Simulation	Launch, transfer, orbit choice, survey, event, transmission, result.
4. NASA Data	CSV parsing, selected values, data display, gameplay modifiers.
5. Visual Polish	Canvas assets, stars, Moon, spacecraft, particles, transitions.
6. Audio	Procedural BGM, UI sounds, launch, survey, transmission, outcomes.
7. Testing	Normal run, invalid design, low power, high science, communication issue, failure, replay.

50. Five-Day MVP Priority

1. CORE GAMEPLAY
2. RESOURCE SYSTEM
3. CHECKPOINT SYSTEM
4. BASIC CANVAS VISUALS
5. NASA DATA
6. FINAL REPORT
7. ANIMATION
8. AUDIO
9. POLISH

Rule for scope

A functioning game must exist before decorative features are added. If a feature threatens the playable MVP, it moves to the post-MVP list.

51. MVP Definition

The MVP is complete when a player can perform the following without developer intervention:

14. Start the game and understand the mission.
15. Select a launch vehicle.
16. Select a power system.
17. Select a scientific instrument.
18. See the resulting mission statistics.
19. Pass or fail a design check.
20. Launch the spacecraft.
21. Choose a lunar orbit.
22. Reach the lunar survey.
23. See genuine NASA-based lunar information.
24. Choose an observation mode.
25. Respond to one mission event.

26. Choose a transmission mode.
27. Complete, partially complete, or fail the mission.
28. Receive a final mission report.
29. Play the mission again.

52. Project Success Criteria

30. A new player understands the game within approximately 30-60 seconds.
31. A complete playthrough takes less than 10 minutes.
32. The player makes meaningful engineering decisions.
33. Decisions alter mission state and later outcomes.
34. The game uses a clear connection to real lunar science.
35. The final report explains how choices changed the outcome.
36. The experience runs reliably in a modern browser.
37. The project is visually and audibly polished enough for a live hackathon demonstration.

53. What Makes LUNA-01 Distinct

- NASA science is part of the gameplay, not just background text.
- The game demonstrates engineering trade-offs through consequences.
- The entire visual world is generated procedurally with code.
- Audio is generated procedurally with Web Audio.
- The whole experience is compact enough to understand in minutes.
- The game is replayable without requiring multiple missions.

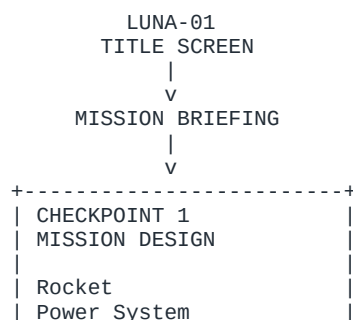
54. Intended Player Experience

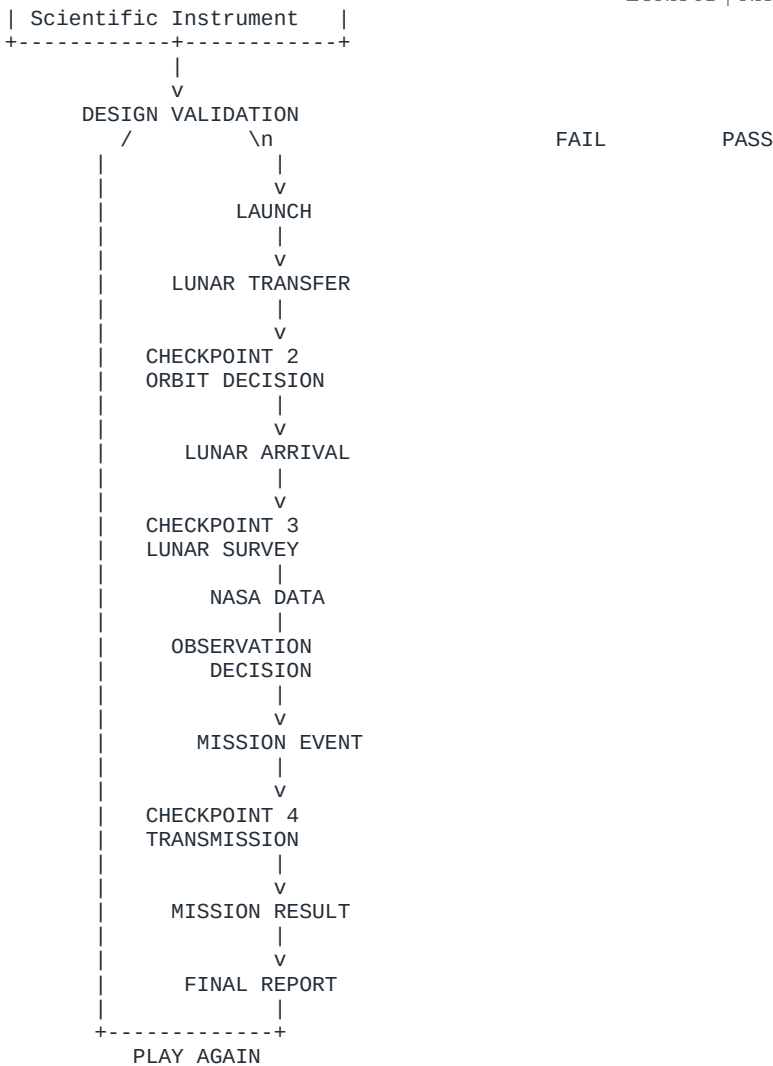
```

START
"I am designing a spacecraft."
  |
DESIGN
"Can I fit everything within the constraints?"
  |
LAUNCH
"My mission is underway."
  |
ORBIT
"Which strategy should I choose?"
  |
SURVEY
"This is real lunar information."
  |
EVENT
"What should I sacrifice?"
  |
TRANSMISSION
"Can I get the data home?"
  |
REPORT
"These were the consequences of my decisions."

```

55. Final Game Flow - Full Blueprint





56. Final Design Statement

LUNA-01: Mission Control is a lightweight, browser-based 2D educational simulation in which players design and operate a single robotic lunar science mission. The player makes a limited number of engineering decisions involving launch vehicle, spacecraft power, scientific instruments, orbital strategy, observation depth, and data transmission. Real NASA lunar environmental information is incorporated into the game so that scientific values influence or explain gameplay. The entire visual world is procedurally rendered with HTML5 Canvas, while CSS handles supporting interface layout and JavaScript controls the simulation, interaction and state. Audio is generated procedurally through the Web Audio API, giving the project a distinctive retro-digital mission-control identity without requiring external sound files.

Core message
Space mission design is a continuous process of balancing objectives, resources, engineering choices, and risk.